

Bao cao do an: He thong tram sac xe dien thong minh

Muc tieu: xay dung mo hinh tram sac xe dien co giam sat, dieu khien va thanh toan qua ung dung.

Thanh phan: ESP32 Station, ESP32 Gateway, Node.js Backend, MySQL, Web Admin va Mobile App Expo.

Giao tiep: Station <-> Gateway bang Modbus RTU/RS485; Gateway <-> Backend bang MQTT; Web/App <-> Backend bang REST API.

Tinh nang: do V/A/W, nhiet do, do am; hien thi QR tren TFT; dieu khien relay 2 cong sac; quan ly user, vi tien, phien sac va nap tien.

Ket qua: he thong hoat dong duoc theo luong IoT hai chieu, co dashboard quan tri, app nguoi dung, co che bao ve qua dong/qua nhiet/mat ket noi.

Xem them: README.md, docs/phu-luc.md va cac hinh trong docs/ket-qua-de